ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE E

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET Mechatronic Systems paper: FIVE questions of 20 marks. Electronic components and waveforms, motor drives and relay control circuits, hydraulic symbols and circuit design, electro-hydraulic control with safety, and systematic fault-finding.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Draw all circuits and symbols neatly, using a ruler. Use standard (ISO / IEC) symbols.
5. Write neatly and legibly.

QUESTION 1

1.1 Define the term diode. (2)

1.2 Name the THREE terminals of a silicon-controlled rectifier (SCR), and explain how the SCR is switched on. (5)

1.3 Draw a fully labelled diagram of a square waveform, clearly showing the amplitude, the period, the high level and the low level. (5)

1.4 Give TWO reasons why a microcontroller, rather than a microprocessor, is used to control a small robot. (4)

1.5 Draw the symbol of a PNP transistor and label its terminals. (4)

[20]

QUESTION 2

2.1 Explain how pulse-width modulation (PWM) controls the speed of a DC motor. (4)

2.2 Name FOUR parts of a stepper motor drive system. (4)

2.3 Explain the difference between open-loop and closed-loop control, and give ONE mechatronic example of each. (4)

2.4 Design a relay control circuit for a DC motor. The circuit must have a START push button, a STOP push button, an emergency stop, a holding (latching) contact and a green lamp that shows when the motor is running. (8)

[20]

QUESTION 3

3.1 Draw the symbol of EACH of the following:

3.1.1 Fixed-displacement hydraulic pump (one direction of flow) (2)

3.1.2 Pressure relief valve (3)

3.1.3 Double-acting cylinder (2)

3.2 State Pascal's law and give ONE application of it in a mechatronic system. (3)

3.3 List FOUR functions of hydraulic oil. (4)

3.4 Design a hydraulic circuit for a workshop vehicle hoist. Make use of: a double-acting cylinder; a manually operated 4/3-way valve with a tandem centre; a pressure relief valve; and a pilot-operated check valve that holds the load in position. (6)

[20]

QUESTION 4

4.1 Explain the function of the solenoid on an electro-hydraulic directional control valve. (2)

4.2 Differentiate between a normally open (NO) and a normally closed (NC) push button, and give ONE use of each in a control circuit. (4)

4.3 A double-acting cylinder is controlled by a 4/2-way valve with a single solenoid (Y1) and a spring return. Design the electro-hydraulic working circuit and the electrical control circuit so that: pressing START (S1) makes the cylinder extend; the cylinder stays extended (relay K1 holds itself) while it moves out; and when the cylinder reaches the end of its stroke, limit switch S2 releases K1 so that the cylinder retracts. (10)

4.4 Give FOUR safety rules to follow when working on a pressurised hydraulic system. (4)

[20]

QUESTION 5

5.1 List, in the correct order, FIVE steps of a systematic fault-finding procedure. (5)

5.2 A hydraulic cylinder extends much more slowly than normal. Give THREE possible causes and the remedy for each. (6)

5.3 Explain how to measure the voltage across a solenoid coil safely with a digital multimeter. (4)

5.4 Explain why overheating of hydraulic oil is a problem, and give TWO possible causes of overheating. (5)

[20]

TOTAL: 100

END OF QUESTION PAPER